Mobile Money User Activity Classification Report

Executive Summary

We built a pipeline to classify mobile money users into low, medium, and high activity segments using SM

Introduction

Mobile money is a primary financial channel in Cameroon. Understanding usage patterns supports segm

Data Collection Methodology

We collected anonymized SMS exports from MobileMoney and OrangeMoney and paired them with parti

Data Cleaning and Preparation

We standardized multilingual headers, removed duplicates, filtered OTP and promotional messages, and

Exploratory Data Analysis

EDA shows a clear separation between low and high activity users driven by transaction volume and freq

Modeling Approach

We trained logistic regression, decision tree, and random forest classifiers. Macro F1 was used due to cla

Results and Interpretation

Top features: total_transaction_volume and activity_score. These capture sustained usage and high-valu

Business Insights and Recommendations

- Reward high-volume users to improve retention.
- Offer merchant bundles for payment-heavy users.
- Use failed attempts to detect friction and improve UX.

Limitations and Ethics

Sample size at the user level is small, and SMS parsing has edge cases. The model should not be used

Conclusion

The project delivers an end-to-end pipeline from SMS ingestion to classification. Expanding the participan